Lab 7: Fuzzing for Input Validation Bugs

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Problem statement: L6_L7_11 — Amusement Park Ride Access

Step 2: Running the buggy processor.py

Command: `python processor.py`. Invalid input `abc` crashes `validate_height_and_fastpass` with an unhandled ValueError.

```
==== Running validate_height_and_fastpass ====
Enter rider spec (Height:FastPass, e.g. 140:YES): abc
Traceback (most recent call last):
  File "C:\Users\mdaga\SE-Lab\Lab-7\processor.py", line 41, in <module>
    main()
    ~~~~^^
  File "C:\Users\mdaga\SE-Lab\Lab-7\processor.py", line 30, in main
    print(validate_height_and_fastpass(r_input))
          ~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "C:\Users\mdaga\SE-Lab\Lab-7\processor.py", line 2, in validate_height_and_fastpass
    height_str, pass_str = rider_spec.strip().split(':')
    ^^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Step 4: Hypothesis tests failing on the buggy code

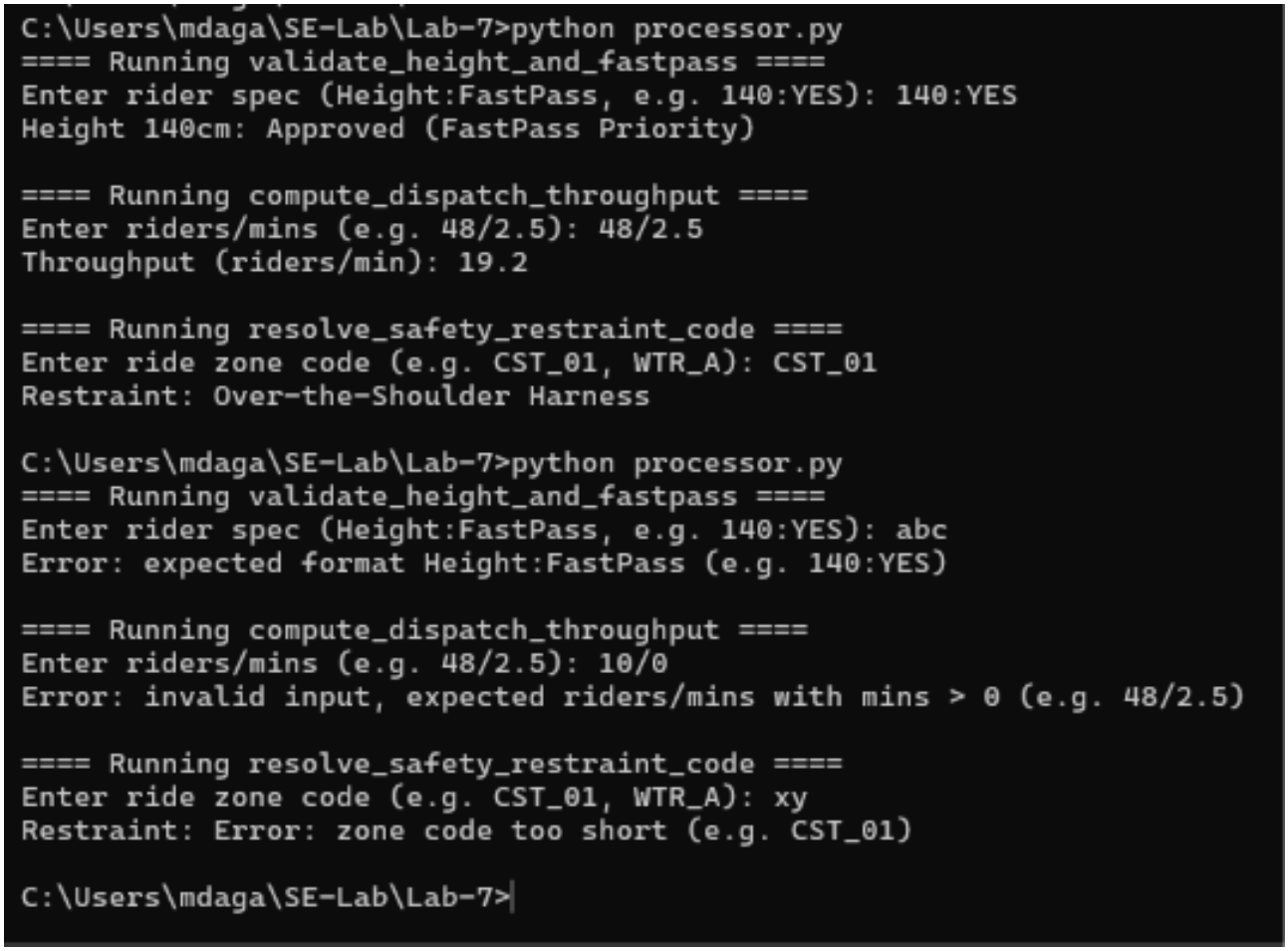
Command: `python -m pytest test_processor.py`. All 3 tests fail; Hypothesis found 4, 7 and 2 distinct failures (13 total) across the three functions. Example shown: `'0:'` raises KeyError and an empty height raises ValueError.

```
=================================================== short test summary info ====================================================
FAILED test_processor.py::test_validate_height_and_fastpass_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
FAILED test_processor.py::test_compute_dispatch_throughput_safe - ExceptionGroup: Hypothesis found 7 distinct failures. (7 sub-exceptions)
FAILED test_processor.py::test_resolve_safety_restraint_code_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
==================================================== 3 failed in 5.58s =====================================================
```

```
=================================================================== FAILURES ===================================================================
_____________________________________ test_validate_height_and_fastpass_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mdaga\SE-Lab\Lab-7\test_processor.py", line 29, in test_validate_height_and_fastpass_no_crash
  |     @given(rider_specs)
  |                ^^^
  | ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mdaga\SE-Lab\Lab-7\test_processor.py", line 32, in test_validate_height_and_fastpass_no_crash
    |     result = validate_height_and_fastpass(s)
    |   File "C:\Users\mdaga\SE-Lab\Lab-7\processor.py", line 5, in validate_height_and_fastpass
    |     has_fastpass = pass_map[pass_str.strip().upper()]
    |                    ~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\mdaga\SE-Lab\Lab-7\test_processor.py", line 35, in test_validate_height_and_fastpass_no_crash
    |     assert False, f"validate_height_and_fastpass crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: validate_height_and_fastpass crashed on '0:' with: KeyError: ''
    | assert False
    | Failing test case: test_validate_height_and_fastpass_no_crash(
    |     s='0:',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\mdaga\SE-Lab\Lab-7\test_processor.py:34
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mdaga\SE-Lab\Lab-7\test_processor.py", line 32, in test_validate_height_and_fastpass_no_crash
    |     result = validate_height_and_fastpass(s)
    |   File "C:\Users\mdaga\SE-Lab\Lab-7\processor.py", line 3, in validate_height_and_fastpass
    |     height = int(height_str)
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
```

Step 5: Running the fixed processor.py

Command: `python processor.py`. Valid inputs give the same results as before; invalid inputs (`abc`, `10/0`, `xy`) now return error messages instead of crashing.



Step 6: Hypothesis tests passing on the fixed code

Command: `python -m pytest test_processor.py`. All 3 tests pass.

